1 **(a)** $45 : 120$ **[M1]**

$3 : 8$ **[A1]**

(b) $3.6 \times 100 = 360$ cm **[M1]**

$\dfrac{360}{2+3+7} \times 7$ **[M1]**

210 **[A1]**

[Total Marks: 5]

2 **(a)** Total parts $= 2 + 3 + 5 = 10$ **[M1]**

Ben's share $= \dfrac{3}{10} \times 840 = 252$ **[A1]**

(b) $\dfrac{5}{10}$ **[M1]**

$\dfrac{1}{2}$ **[A1]**

[Total Marks: 4]

3 **(a)** $16.8 \times 250\,000 = 4\,200\,000$ cm **[M1]**

42 **[A1]**

(b) Scale of area $= 1^2 : (2.5)^2$ OR 1 cm^2 represents 6.25 km^2 **[M1]**

8.96 **[A1]**

(c) 14 cm^2 represents 56 km^2 $\implies$ 1 cm^2 represents 4 km^2 **[M1]**

Linear ratio $= \sqrt{1}$ cm represents $\sqrt{4}$ km $\implies$ 1 cm to 2 km **[M1]**

$200\,000$ **[A1]**

[Total Marks: 7]

4 **(a)** Alex : Ben $= 15 : 20$ and Ben : Chris $= 20 : 24$ **[M1]**

$15 : 20 : 24$ **[A1]**

(b) Difference in parts $= 24 - 15 = 9$ parts $\implies$ 9 parts $= 16\,200$ **[M1]**

Total parts $= 15 + 20 + 24 = 59$ **[M1]**

$P = \dfrac{16\,200}{9} \times 59 = 106\,200$ **[A1]**

(c) Alex's share $= \dfrac{15}{59} \times 106\,200 = 27\,000$ **[B1]**

$27\,000 \times \left(1 + \dfrac{4.2}{100}\right)^5$ **[M1]**

$33\,167$ **[A1]**

[Total Marks: 8]

5 **(a)** Actual distance $= 8.4 \times 250\,000 = 2\,100\,000$ cm **[M1]**

Distance in m $= 21\,000$ or distance in km $= \dfrac{2\,100\,000}{100\,000}$ **[M1]**

21 **[A1]**

(b) Area scale factor $= 1^2 : (250\,000)^2 = 1 : 6.25 \times 10^{10}$ **[M1]**

35 km$^2 = 35 \times 10^{10}$ cm^2 **[M1]**

Map area $= \dfrac{35 \times 10^{10}}{6.25 \times 10^{10}} = 5.6$ **[A1]**

[Total Marks: 6]

6 **(a)** Let total workers be $8x$. Males $= 5x$, Females $= 3x$. **[M1]**

Full-time males $= \dfrac{3}{5} \times 5x = 3x$. Part-time males $= \dfrac{2}{5} \times 5x = 2x$. **[M1]**

Full-time females $= \dfrac{4}{5} \times 3x = 2.4x$. Part-time females $= \dfrac{1}{5} \times 3x = 0.6x$. **[M1]**

Total full-time $= 3x + 2.4x = 5.4x$. Total part-time $= 2x + 0.6x = 2.6x$. Ratio is $5.4 : 2.6 = 27 : 13$. **[A1]**

(b) Difference $= 3x - 0.6x = 2.4x$ **[M1]**

$2.4x = 42 \implies x = 17.5$ **[M1]**

Total workers $= 8x = 8 \times 17.5 = 140$ **[A1]**

[Total Marks: 7]